Project Proposal & Literature Review

Building and Validating a Lightweight No-Reference Image-Quality Metric (Project 28)

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Abstract

Most images we actually look at have been compressed, resized, or filtered before we see them, and in those settings the clean original is usually gone. PSNR, SSIM and LPIPS all need that original, so they cannot score a photo that arrives through a messaging app or a streaming service. This project builds a no-reference quality metric: a small regression head trained on features from a frozen pretrained network, with KADID-10k as the source of human labels (10,125 distorted images with DMOS scores). We have already reproduced the reference baselines on a content-based split (seed 42). LPIPS reaches SROCC 0.808 against human scores, SSIM 0.627 and PSNR 0.667; those numbers are the target our model has to approach without seeing the reference. Code and results are public in our GitHub repository.

1 Problem Statement & Motivation

Take a single image that has been compressed, blurred, or re-touched, with no clean version available, and try to say how good it looks in a way that matches human judgement. That is the no-reference image quality assessment (NR-IQA) problem. The metrics we use in the course, PSNR and SSIM, compare a test image against a reference and were designed for controlled pipelines where that reference exists. In practice it often does not: photos are uploaded, re-encoded, enhanced by apps, or generated. Recent work also shows that full-reference scores can be pushed up without a real gain in perceived quality (Siniukov et al., 2022; Chen et al., 2025), so the interesting question is how far we can get from the distorted image alone. Our project answers it with a small learned model, validated against human opinion scores.

2 Related Work

Work on NR-IQA has moved in two directions over the past few years. The first reuses a large pretrained model and adds something small on top. CLIP-IQA scores an image by comparing its CLIP embedding to antonym prompts such as "a good photo" and "a bad photo" (Wang et al., 2023); MA-CLIP later showed that the magnitude of the CLIP feature is itself a useful quality cue (Liao et al., 2026). ARNIQA keeps the same economy with a different backbone: a frozen encoder learns a distortion manifold and quality is read out by a linear regressor (Agnolucci et al., 2024). A recent WACV study compares six frozen backbones under an identical small head and finds that even the choice of activation function matters (Yadav et al., 2026). The second direction builds lightweight end-to-end models; LAR-IQA uses MobileNetV3 branches and reports KADID-to-TID2013 transfer

(Jamshidi Avanaki et al., 2024). Surveys (Ma et al., 2025) and the newest full-reference work (Chen et al., 2025) agree that generalization, not in-dataset accuracy, is the unsolved part.

Our plan follows the first direction and keeps the adaptation as small as possible. We use no language prompts, and unlike Jamshidi Avanaki et al. (2024) we do not train the encoder: a small frozen backbone supplies features, a two-layer head maps them to a score, and evaluation uses a content split plus a cross-dataset check. Table 1 lists the references this plan builds on.

Table 1: Key references.

Reference	Year	Relevance to this project
(Zhang et al., 2018)	2018	Learned FR perceptual metric; strongest baseline to approach
(Ding et al., 2022)	2022	Structure + texture FR metric; optional baseline
(Siniukov et al., 2022)	2022	FR metrics can be gamed; motivates the NR framing
(Wang et al., 2023)	2023	Frozen CLIP prompts for NR-IQA; frozen-backbone philosophy
(Agnolucci et al., 2024)	2024	Frozen encoder + linear regressor; closest design to ours
(Jamshidi Avanaki et al., 2024)	2024	Lightweight mobile NR-IQA; KADID→TID2013 protocol
(Chen et al., 2025)	2025	Limits of the FR paradigm; motivation
(Ma et al., 2025)	2025	Recent survey; structures this review
(Lao and Yu, 2025)	2025	Hybrid semantic + low-level metric; recent point of reference
(Yadav et al., 2026)	2026	Controlled frozen-backbone study; head design matters
(Liao et al., 2026)	2026	Training-free CLIP-magnitude metric; freshest method

3 Dataset & Baseline Plan

Dataset. We use KADID-10k (Lin et al., 2019). It contains 81 pristine images, each degraded by 25 distortion types at five severity levels, giving 10,125 distorted images. Every image has a differential mean opinion score (DMOS) between 1 and 5, higher meaning better perceived quality. The database is free for research use and we cite it as required. We keep all 10,125 images but split by reference image: 65 references for training, 16 for testing, with seed 42 fixed before any experiment. Splitting by distorted image instead would put different versions of the same content on both sides and inflate every correlation, so we avoid it deliberately. TID2013 (3,000 images, 24 distortion types) is held back for the M3 cross-dataset check. The archive we downloaded is recorded byte-for-byte (3,067,408,471 bytes), and the SHA-256 of the label file matches the authors' mirror, so a rerun on another machine starts from the same data.

Baselines. We compare against PSNR and SSIM, computed on RGB because KADID contains colour distortions that a grayscale comparison would miss, and against the pretrained LPIPS metric (Zhang et al., 2018) with its AlexNet backbone. Correlations follow the usual IQA convention:

SROCC directly, PLCC and RMSE after a four-parameter logistic fit, because metric outputs and DMOS are not on the same scale. Sixty-four test pairs are (near-)identical to their reference and would give an infinite PSNR, so PSNR is capped at 60 dB and those pairs are counted. Table 3 shows the baseline numbers on the 2,000 test images; these are what our no-reference model has to match using only the distorted image.

Table 2: Dataset summary.

Dataset	**Subset used**	**Link**
KADID-10k	10,125 distorted + 81 refs; 65/16 split	KADID-10k page
TID2013	3,000 distorted + 25 refs; M3 only	TID2013 page

Table 3: Preliminary baseline correlations on the M1 test split (2,000 distorted images), computed by our own pipeline.

Metric	**SROCC**	**PLCC**	**RMSE**
PSNR	0.667	0.670	0.803
SSIM	0.627	0.616	0.852
LPIPS (Zhang et al., 2018)	0.808	0.807	0.639

4 Proposed Method

The pipeline has five steps.

1. **Input.** The distorted image alone, resized to a fixed working resolution, with horizontal flips for augmentation.
2. **Backbone.** MobileNetV3-Large (about 5.4M parameters, ImageNet weights) produces a pooled feature vector. We never fine-tune it, so features are computed once per image and reused.
3. **Head.** A two-layer MLP maps the feature vector to one score. This is the only part we train. Keeping it small is deliberate: with 8,125 training images a larger head would memorize them.
4. **Training.** MSE against DMOS, Adam, early stopping, seed 42. A pairwise ranking term is a stretch item.
5. **Evaluation.** The same split and metrics as the baselines, plus KADID→TID2013 transfer at M3.

The contribution is not a new network. We take the frozen-feature approach and keep it as small as possible, with a split that avoids leakage and a generalization check that most in-dataset comparisons skip.

5 Feasibility & Compute Plan

Compute tier: **CPU-OK / GPU-light**. The course machine (8-core CPU, 16 GB RAM) can run every step; we also use a free Colab T4 to speed up feature extraction. Extracting features for

10,125 images takes minutes on the T4 and a few hours on CPU, but it happens once; after that, training the head takes seconds to minutes. The baseline table in this report already ran end-to-end on the free Colab tier in under ten minutes. Checkpoints and result files are stored in the repository and in a private drive, and every script is idempotent with a fixed seed, so losing a Colab session costs time, not results.

5.1 Minimum Viable Deliverable (MVP)

A frozen MobileNetV3-Large plus a trained MLP head that predicts DMOS on the 16 test references and is reported in the same table as the baselines. If it works, we have a valid comparison; if the correlation is low, the per-distortion breakdown still tells us where the frozen features fail.

5.2 Stretch Goal

Train on KADID and test on TID2013, add one ablation (backbone size or head width, optionally the ranking loss), and if time allows compare with MA-CLIP's released code as a training-free reference point (Liao et al., 2026).

6 Risk Assessment

- **Overfitting.** A frozen encoder limits the damage, but the head can still memorize content. We keep it small, stop early on a validation slice, and report the curve rather than only the best epoch.
- **Some distortion types may be poorly represented in MobileNet features**, especially the colour distortions. If the per-type breakdown shows a weak family, we switch to EfficientNet-B0 or a small ViT; the fallback is a compact CNN trained from scratch.
- **Colab sessions die without warning.** Checkpoints and results are copied to Drive and to the repository, and the pipeline reruns from a fixed seed, so a lost session costs a few minutes of recomputation.
- **KADID is research-use only.** We do not redistribute the images; the repository contains code and derived numbers, not the dataset.

7 Work Breakdown

AI-Tool Usage Disclosure

AI-tool disclosure: code scaffolding, literature-search assistance, and LaTeX/prose editing were AI-assisted; all experiments were run and verified by the team, who understand and can explain the full pipeline.

References

Maksim Siniukov, Dmitriy Kulikov, and Dmitriy Vatolin. Applicability limitations of differentiable full-reference image-quality metrics, 2022.

Table 4: Milestone task allocation (see also the course project catalog entry for this project).

Milestone	**Student A**	**Student B**	**Student C**
M1 (Wk 3)	Dataset pipeline and integrity checks; baseline correlations (done); literature table	Backbone and head design; feasibility test	Evaluation protocol and harness (done); repository hygiene
M2 (Wk 6)	Full baseline table; per-distortion analysis	First trained checkpoint; backbone ablation	Preliminary comparison table; figures
M3 (Wk 9)	Final baselines; TID2013 pipeline	Final model and ablations; ranking-loss variant	Cross-dataset table; demo; report figures

Du Chen, Tianhe Wu, Kede Ma, and Lei Zhang. Toward generalized image quality assessment: Relaxing the perfect reference quality assumption. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12742–12752, 2025.

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Ankit Yadav, Ta Duc Huy, and Lingqiao Liu. Revisiting vision–language foundations for no-reference image quality assessment. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 5416–5425, 2026.

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Zhiqiang Lao and Heather Yu. Scene perceived image perceptual score (SPIPS): Combining global and local perception for image quality assessment, 2025.

Hanhe Lin, Vlad Hosu, and Dietmar Saupe. KADID-10k: A large-scale artificially distorted IQA database. In *2019 Tenth International Conference on Quality of Multimedia Experience (QoMEX)*, pages 1–3. IEEE, 2019.